

HANDA: A Multi-Hazard Early Warning and Community Preparedness System for the Philippines.

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Course: BS-Nursing 4D

Date: August 20, 2026

I. Introduction

Natural disasters are major adverse events resulting from natural processes of the Earth, including earthquakes, landslides, tropical cyclones, floods, volcanic eruptions, and other hazards. These events can result in deaths, injuries, displacement, infrastructure damage, and disruption of livelihoods. Globally, approximately 10,000 to 20,000 people die from disasters each year, while more than 100 million people are affected annually. However, disaster-related deaths can be reduced through better forecasting, early warning systems, resilient infrastructure, and coordinated community responses (Ritchie et al., 2022).

The Philippines is particularly vulnerable to natural disasters because of its geographical location and exposure to tropical cyclones, floods, landslides, earthquakes, and volcanic hazards. The country has therefore developed various disaster risk reduction and early warning initiatives involving agencies such as the Philippine Atmospheric, Geophysical and Astronomical Services Administration (PAGASA), the Philippine Institute of Volcanology and Seismology (PHIVOLCS), the National Disaster Risk Reduction and Management Council (NDRRMC), and local government units. Project UP NOAH, originally established by the Department of Science and Technology (DOST) in 2012, provides open-access hazard information, including flood, landslide, and storm-surge maps, to support disaster preparedness and risk assessment.

Despite these existing systems, disaster information does not always translate into immediate and appropriate action at the community level. Research on Philippine communities has identified continuing challenges involving communication, public awareness, capability building, and the accessibility and usefulness of warnings (Gumiran & Daag, 2021; Pelone & Arellano, 2024).

II. Problem Description

Natural disasters remain a significant problem in the Philippines because hazards can occur rapidly and affect communities differently depending on their location and vulnerability. A warning that simply informs residents that a typhoon, flood, or landslide is approaching may not sufficiently explain what people should do, where they should evacuate, or how urgently they need to respond.

Studies emphasize that an effective early warning system requires more than hazard detection. It must integrate risk knowledge, monitoring and warning services, communication and dissemination, and the capacity of communities to respond (Gumiran & Daag, 2021). In Philippine landslide-prone communities, Gumiran and Daag (2021) found that a people-centered and multi-stakeholder approach can strengthen the sustainability and effectiveness of early warning systems.

Similarly, a 2024 study in Tagum City, Davao del Norte, found that households demonstrated generally good flood preparedness but still had deficiencies in capability building and information distribution. The researchers recommended strengthening training and improving the dissemination of early warning information (Pelone & Arellano, 2024).

These concerns demonstrate the need for a technology-enabled system that does not merely provide disaster information but converts official hazard data into localized, understandable, timely, and actionable instructions for residents.

III. Proposal Solution

The proposed solution is HANDA, a mobile and web-based multi-hazard early warning and community preparedness system designed for Philippine communities. HANDA would integrate official information from agencies such as PAGASA and PHIVOLCS with location-based hazard information from resources such as Project UP NOAH. Its primary purpose is to provide residents with timely warnings and clear instructions before, during, and after a disaster.

Project UP NOAH currently provides open-source and accessible scientific disaster data, including flood, landslide, and storm-surge hazard maps. Its platform allows users to enter their location and view the corresponding hazard risks in their surroundings. HANDA would build upon this concept by presenting the information through a simpler, community-centered interface.

HANDA would also provide location-based disaster alerts using GPS or a manually selected barangay, allowing residents to receive timely warnings through push notifications or SMS when their area is threatened. It would feature a multi-hazard dashboard covering tropical cyclones, floods, landslides, earthquakes, volcanic eruptions, and storm surges, with simple risk levels such as Normal, Monitor, Prepare, Evacuate, and Emergency. Additionally, HANDA provides action-based warnings, transforming technical hazard information into clear instructions that residents can immediately understand and follow. This supports the use of impact-based forecasting, where warnings focus not only on the hazard itself but also on its possible effects on communities.

The system would also include an interactive hazard map showing areas vulnerable to floods, landslides, and storm surges, as well as important locations such as evacuation centers, hospitals, fire stations, and police stations. During emergencies, its evacuation assistance feature would help users locate nearby designated evacuation centers and provide directions and information about their activation status. Because disasters can disrupt electricity and internet connectivity, HANDA would provide offline emergency information, including emergency contacts, evacuation procedures, first-aid instructions, and previously downloaded hazard maps. A community reporting feature would allow residents to report flooding, landslides, blocked roads, and damaged infrastructure using photographs and location information, while an LGU and barangay dashboard would allow disaster personnel to monitor warnings, community reports, affected areas, and evacuation activities. Overall, HANDA would integrate official hazard information, location-based alerts, mapping, evacuation support, community participation, and LGU coordination into a single platform, making disaster information more timely, accessible, localized, understandable, and actionable.

IV. Conclusion

Natural disasters cannot always be prevented, but their impacts can be reduced through timely information, preparedness, effective communication, and coordinated response. The Philippines already possesses important disaster-monitoring and hazard-mapping capabilities through agencies such as PAGASA, PHIVOLCS, and Project UP NOAH. However, there remains a need to make disaster information more localized, understandable, accessible, and actionable for communities.

HANDA proposes a technology-enabled solution that integrates official hazard information, location-based alerts, interactive hazard maps, evacuation assistance, offline emergency resources, community reporting, and LGU coordination into one platform. By combining scientific information with community participation, HANDA can strengthen the connection between hazard detection and actual community action. The proposed system therefore offers a practical and scalable approach toward improving disaster preparedness and resilience in Philippine communities.

V. References

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